

Generalization Improvements for Musical Audio Effect Classification

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<https://github.com/NonabstractPhoton/AudioEffectMulticlassifier>

Abstract—Audio effects are integral to the sound profile of electric instruments within real-world use. The presence or absence of key effects broadly characterizes different styles of play. However, despite their ubiquity, audio effects have proven notoriously difficult to classify, especially in the multiclass case. In this work, we improve upon the best known architectures for neural network based classifiers while developing a vast dataset to support broad generalization. In doing so, we develop models that are comparable to the best known results in the single label task while making headway into the largely unexplored multiclass problem.

I. Introduction

All musicians that utilize electric instruments —such as the electric guitar, bass guitar, or synthesizers —apply combinations of audio effects to their sound. These effects can be applied in post-production, however it is more common for them to be applied live through signal processing utilities. Some of these are built into the instrument or amplifier, while others are external, and broadly referred to as "pedals" due to their primary form factor. Effect suites are typically comprised of subtle, contextual effects —such as compressors, band pass filters, and tone knobs —as well as distinctive, genre defining effects, such as overdrive, distortion, and flanger. The former are typically handled by inbuilt utilities, whereas the latter are overwhelmingly applied through pedals. It is these pedal effects that are most valuable to classify, as they largely characterize unique sounds. In this work, we aim to classify audio based on the presence of one or two of the following effects: reverb, chorus, flanger, bitcrusher, phaser, tremolo, distortion, and overdrive. Including no effect, this gives us 9 classes. Many prior works have focused on the single-class problem in audio effect classification, however comparatively little progress has been made on the multiclass problem. Furthermore, although existing approaches have shown promise, audio effect classification has yet to mature into a thoroughly explored field. Thus, this work aims to grow that body of knowledge and contribute to the development of accurate, reliable, and ultimately useful audio effect classifiers.

II. Related Works

A hallmark work in the broader task of audio classification found that, among several different architectures, a convolutional neural network operating on both the Mel-spectrogram and waveform of the audio data performs best in the single-class case [1]. This general fact can be seen in the architecture choices of many works that address the effect classification problem. The exact best-performing architecture from that initial work was later applied to the electric instrument effect classification problem with reasonable success [2], however it is clear that extensions to a purely convolutional basis generally perform better [3], [4].

A key pitfall in existing works is their lack of generalizability. Even works that offer state of the art performance on samples similar to their training data struggle with test sets sourced from other datasets [4], likely due to differences in clean sound and the internal variations in how effects with the same name may actually be applied. Therefore, utilizing a sufficiently broad training set with several different audio sources seems key to creating more generalizable classifier.

Another notable issue that stems from this fact is the lack of meaningful benchmarks. Since the field is still developing, there are only a few datasets that see frequent use. Although those datasets are excellent, they offer limited internal variety in terms of the factors that would make a model generalizable to new sounds. Consequently, this means that most test sets lack internal variety, and thus only demonstrate generalizability to a first order sense if they have been intentionally chosen to be from a different source than the train set. Ideally, as the field matures, definitive and widespread test sets created with internal variety in mind will become standardized across works. However, as it currently stands, limited organic data and variation are available to researchers exploring the task of effect classification.

III. Model Architectures

In this work, we develop two new classification model architectures that both extend from the so-called Wavegram-Logmel-CNN [1]. Our primary mode of extension is through the addition of transformer blocks, which allows us to make use of the powerful multi-head attention mechanism [5]. For one of our tested architectures, we add three transformer blocks before the final fully-connected layer of the best Wavegram-Logmel-CNN architecture from the original work. Besides the addition of these blocks, the original architecture remains largely intact. Our other architecture, meanwhile, forgoes many of the larger convolutional layers in the Wavegram-Logmel-CNN in favor of a primarily transformer-block oriented approach. We will henceforth refer to these two architectures as the CNN Extended and CNN + Transformer respectively.

In brief, the Wavegram-Logmel-CNN is a convolutional neural network that operates on a novel embedding of the audio. This embedding involves concatenating the standard Mel Spectrogram with a Wavegram, which is the result of applying several convolutional layers to a waveform in order to create a sort of learnable Fourier Transform [1]. This embedding formulation is common to both our CNN Extended and CNN + Transformer. The subsequent operations, however, diverge between the two.

- 1) For the CNN + Transformer, this embedding is passed through a series of transformer blocks equal to the number of classes, and then finally to a fully connected projection layer. This final output may be passed through a sigmoid function in order to extract estimated probabilities, however the model omits an explicit sigmoid in order to use a more numerically stable version of binary cross-entropy as its loss function.
- 2) For the CNN Extended, this embedding passes through several additional convolutional layers, as per the best performing architecture from the corresponding paper, and then finally through 3 more transformer blocks before it is passed to the projection layer. This architecture uses the same loss function as the last, and therefore also omits the explicit sigmoid function for the aforementioned reasons.

IV. Dataset Construction

The most notable feature of this work as compared to existing literature comes from its dataset. In order to address the sonic variety problem we discussed earlier, our composite dataset contains several major datasets that were recorded with different equipment, as well as

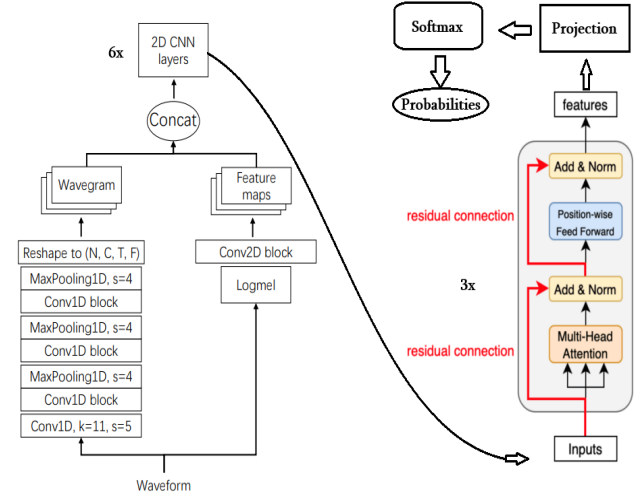


Figure 1: Extended CNN Architecture (176 M)

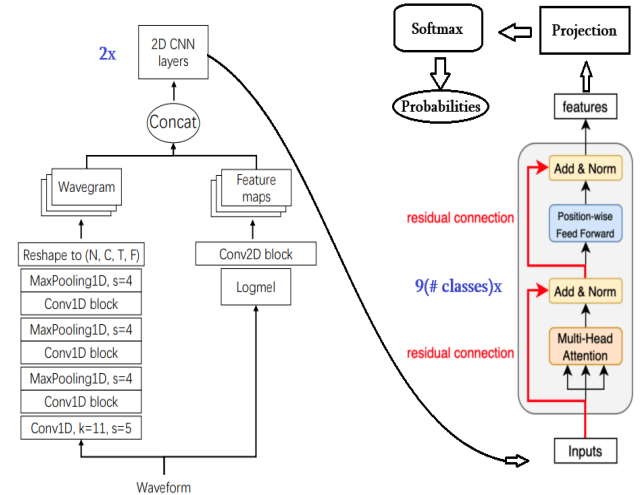


Figure 2: CNN + Transformer Architecture (514 M)

original samples recorded in a variety of different conditions. The publicly available basis datasets include the IDMT-SMT-Audio Effects set [6], the IDMT-SMT-Guitar set [7], the EGFX set [8], and the Guitar TECHS set [9]. The custom samples, meanwhile, include live recordings of an electric guitar and bass guitar with varying EQ and tone settings recorded through both direct input and multiple different microphones. The custom segment also includes computer generated samples using several different electric guitar and bass guitar soundfonts playing both single notes and full pieces. These samples were generated through the Musecore software.

This large basis is then significantly augmented through the application of every audio effect as well

as their combinatorial pairs to all of the clean, i.e. unaffected, audio. With 9 classes, including no effect, this results in a $37\times$ increase in the amount of samples we can extract from clean audio data. These sound effects were applied with Spotify Pedalboard [10] and SoX, the open source Sound eXchange utility.

In order prepare audio for use in the dataset, we split all files into 2 second clips, using a 16-bit integer PCM wav file encoding with a sampling rate of 44.1kHz. We use monophonic continuous audio formatting for maximum generalizability, and convert audio to this format whenever necessary. Upon reformatting all of our samples, including those generated by applying effects to clean samples, we then pack them into hdf5 [11] files based on their effect label in order to reduce overhead during the sampling process. Each sample within these files is indexed, and their indices are stored in separate hdf5 files.

In total, our training dataset includes 242 GB of audio data when packed into hdf5 files. Originally, our dataset included triple-class labels, and contained 496 GB of data. However, after some testing, we opted to reduce the maximum simultaneous effects to 2 due to concerns over realism and the training time tradeoff. Since our selection of pedal effects are far more dramatic than the inbuilt or contextual effects we discussed earlier, it is rare for more than two of them to be used at a time to noticeable degrees. Thus, we determined that pairs are sufficient for practical use of a classifier.

With an estimated 760 hours of audio from many different sources and setups, our dataset is uniquely well equipped to handle broad generalizations to new sound profiles. Not only were several different instruments, settings, and recording media used in the samples, but also many different sources of signal processors for applying the effects. We postulate that the reason many past models have struggled to generalize to new datasets is not only due to difference in clean audio profile, but also differences in what a specific may sound like between signal processors. This gives our composite dataset an out-sized advantage beyond its additional size, as we incorporate several datasets that use different signal processors, and as well as multiple effect sources in our custom samples and augmentation process.

The conjunction of these advantages gives us a dataset suited to training or testing models for broad generalization to multiple sound profiles and audio effect styles.

V. Computing Resources

Both the CNN Extended and CNN + Transformer models were trained across 16 NVIDIA A100 40 GB

GPUs using distributed data parallelism. Training was conducted on the NERSC Perlmutter system across 4 GPU compute nodes. We selected the AdamW optimizer with a batch size of 96 and an initial learning rate of 10^{-5} , with occasional large increases in order to escape local minimums. We also trained in BF16 mixed precision in order to reduce computational time and memory footprint during the forward passes.

VI. Results

For our test set, we used EGSet12 [12] as our clean audio source, and subsequently applied post-processing and audio effects as per the methodologies discussed in the dataset construction section. This resulted in a test set containing roughly 3.9 hours of audio and 1.2 GB of packed waveform data. The following are our test results for both architectures on the full multi-class problem and on the restricted single-class problem.

Table 1. CNN Extended Accuracy

Effect	Multi-Class	Single Class
Reverb	.2541	.2506
Chorus	.7711	.8929
Flanger	.7859	.8929
Bitcrusher	.7859	.8929
Phaser	.7859	.8929
Tremolo	.7859	.8929
Distortion	.8122	.8929
Overdrive	.7866	.8929
No Effect	.9997	.9988
Cumulative	.7519	.8333

Table 2. CNN + Transformer Accuracy

Effect	Multi-Class	Single Class
Reverb	.7911	.8524
Chorus	.7920	.8843
Flanger	.7859	.8929
Bitcrusher	.7859	.8924
Phaser	.7859	.8929
Tremolo	.7859	.8929
Distortion	.8757	.9138
Overdrive	.8018	.8906
No Effect	.9994	.9977
Cumulative	.8226	.9011

The CNN Extended architecture achieved strong, consistent results for all labels in the single class problem with the exception of reverb. We hypothesize this is due to the highly temporal nature of the effect, which is harder to capture through frequency embeddings and the architecture of this network. Comparatively, the CNN extended struggled with the multi-class problem, however this was somewhat expected. Both our results and observations from the literature [2] point to the fact that primarily convolutional architectures struggles to classify multiple audio effects.

The CNN + Transformer architecture also achieved strong results for the single-class problem. Furthermore, unlike the CNN Extended, it effectively identified the reverb effect—likely due to its superior handling of the time series data in the wavegram embedding. The CNN + Transformer also performed well on the multi-class problem, generally out-competing the results of the CNN Extended on the same dataset. Generally, the CNN + Transformer had an accuracy comparable or better than the CNN Extended for most effects in both multiclass and single class problems.

Compared to the basis architecture for these models trained on a highly varied dataset [2], the CNN + Transformer model performed consistently better in single class effect detection accuracy. The CNN Extended also demonstrated superior accuracy, with the exception of the reverb effect. CNN Extended performed the most consistently and the best for single class effects excluding reverb. With this exclusion, CNN Extended outperformed both the CNN + Transformer and the basis architecture for single class.

VII. Conclusion

Both architectures achieved strong generalization results to a dataset with recording conditions largely different from any of the samples they were trained on. Additionally, the CNN + Transformer architecture achieved particularly strong results, competitive with even the most state of the art models [4] on the single-class problem. This bodes well for the performance of future hybrid models in the multiclass problem, as there seems to be a clear advantage in utilizing the combined strengths of convolutional and transformer based architectures.

Nevertheless, despite our optimism, we must highlight a key issue in the audio effect classification paradigm. As we have painstakingly discussed, generalizability is key to the development of an effective classifier. To that end, we constructed a dataset with the maximal possible variety, used an entirely distinct test set, and attempted to make headway into the underexplored multi-class problem. However, the data available

to us, and to all other works in the field thus far, is ultimately lacking in some forms of necessary variety. Although we have incorporated as much attention to variety in recording conditions as possible, our breadth of variation in effect application was unfortunately limited. Despite containing every distinct audio effect dataset we had access to, our data was still ultimately dominated by post-processed, digital effects applied as part of an augmentation process. This pattern is shared by all other major works in the field, as recording style variety remains far easier to source than effect variety. For that reason, our work focused primarily on generalizability through the lens of recording style. Nevertheless, we would like to draw attention to the other side of generalizability argument in the hopes that future developments will make accounting for that source of variety more feasible.

The primary goal of this work was to advance the generalizability and thereby effectivity of audio effect classifiers. Through building upon the excellent contributions of previous works, we have successfully been able to achieve this, especially in regard to variations in recording conditions and addressing the multi-class problem. We now hope that future works will be able build upon these considerations and results in order to eventually create truly generalized audio effect classifiers.

Contributions and Acknowledgments

Atharv Thakur worked on the dataset augmentation process, technical setup for training, and final testing. He also developed and trained the CNN + Transformer architecture.

Winnie Tang worked on developing and training the CNN Extended architecture. She also handled research into existing methodologies and the production of presentation materials as per our obligations to the Rutgers Distributed Deep Learning Systems class.

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